BME 105: Genetics in the Genomics Era

Lecture 7: Sex-determination and dosage compensation

Instructor: Dr. Shelbi Russell,
Assistant Professor

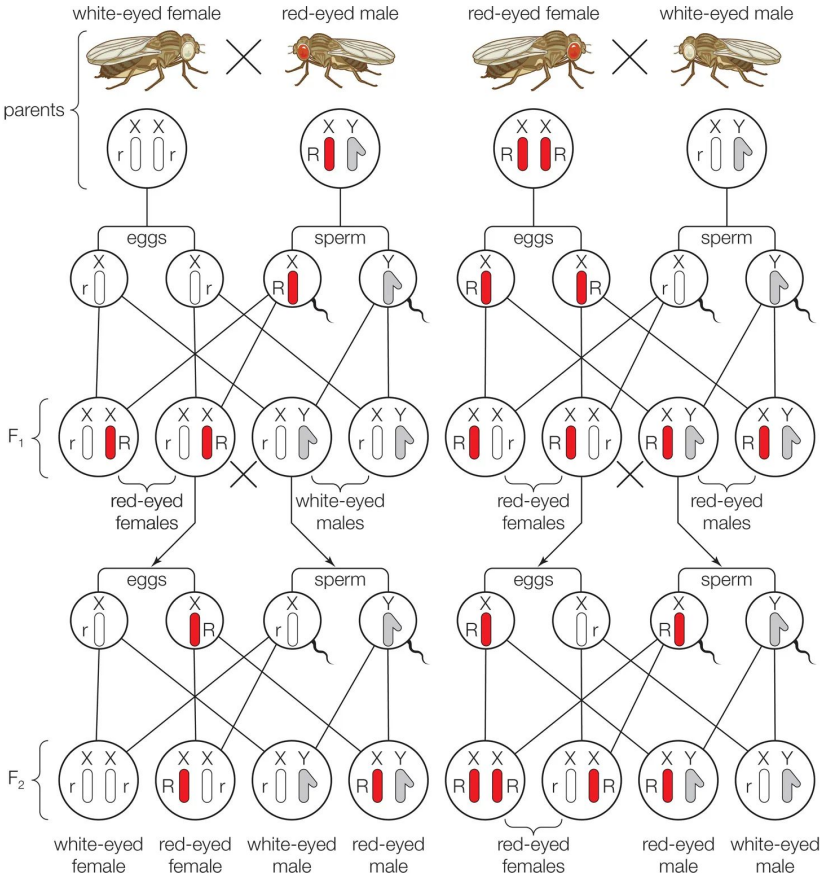
Course website on canvas.ucsc.edu

Some slides modified from Angela Brooks

Lecture 7 Learning Objectives

- Revisit X-linked trait inheritance to be sure you understand enough to:
  - Identify dominant and recessive X-linked traits in pedigrees
  - Calculate the probability of X-linked trait inheritance
- Understand how sex chromosomes are paired during meiosis
  - Be able to describe the outcomes of sex chromosome non-disjunction
    - Know which are viable and/or fertile
- Understand how sex chromosomes determine male and female phenotypes
  - Know how the mechanisms compare between mammals and flies
- Understand sex chromosome <u>dosage compensation</u> in terms of:
  - Why their unbalanced number - compared to autosomes and between sexes - is a problem
  - How the dosage problem is compensated for in mammals, flies, and nematodes (3 ways)
  - How random X-inactivation in mammals produces variable female phenotypes

Chromosomal sex-linked inheritance was discovered in *Drosophila*



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This work confirmed the chromosomal inheritance of genetic information + linkage

Chromosomal theory of inheritance → Nobel Prize in 1933

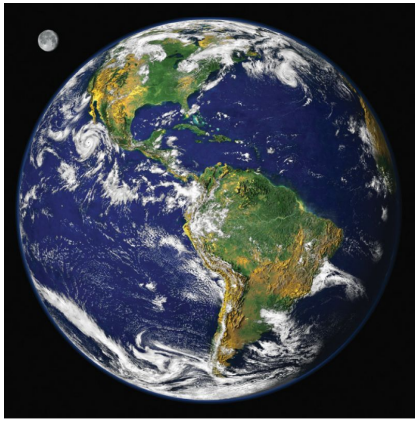
Morgan 1910 Sex-limited inheritance in *Drosophila. Science.*

Morgan, Thomas Hunt, et. al. 1915 The mechanism of Mendelian heredity. *New York: Henry Holt and Co.*

*** Remember, the structure of DNA wasn't known until ~40 years later (Watson and Crick 1953 Nature)*

Human X-linked traits: foundational to chromosomal theory of inheritance

In 1911, E. B. Wilson, a contributor to the chromosome theory of inheritance, combined family studies of the inheritance of color blindness with recent knowledge of the roles of the X and Y chromosomes in sex determination to make the first assignment of a human gene to a particular chromosome. The gene for red-green color blindness, he said, lies on the X because the condition usually passes from a maternal grandfather through an unaffected carrier mother to roughly 50% of the grandsons.



(a)

(b)

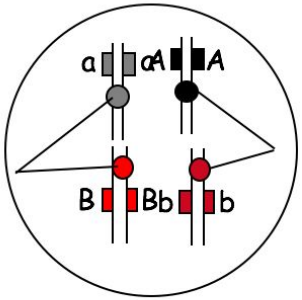
Sutherland-Haan syndrome
Cognitive function, social
Mental retardation, nonspecific
Menkes disease
Occipital horn syndrome
Cutis laxa, neonatal
FG syndrome
Immunodeficiency, moderate and severe
Miles-Carpenter syndrome
Charcot-Marie-Tooth neuropathy, dominant
Mental retardation
X-inactivation center
Premature ovarian failure
Arts syndrome
Cleft palate and/or ankyloglossia
Megalocornea
Epilepsy (Juberg-Hellman syndrome)
Pelizaeus-Merzbacher disease
Spastic paraplegia
Alport syndrome
Cowchock syndrome
Hypertrichosis, congenital generalized
Ptosis, hereditary congenital
Apoptosis inhibitor
Panhypopituitarism
Thoracoabdominal syndrome
Simpson-Golabi-Behmel syndrome, type 1
Split hand/foot malformation, type 2
Hypoparathyroidism
Mental retardation, Shashi type
Lesch-Nyhan syndrome
HPRT-related gout
Lowe syndrome
Borjeson-Forssman-Lehmann syndrome
Testicular germ cell tumor
Hemophilia B
Warfarin sensitivity
Osseous dysplasia (male lethal), digital
Adrenoleukodystrophy
Adrenomyeloneuropathy
Colorblindness, blue monochromatic
Cardiac valvular dysplasia
Emery-Dreifuss muscular dystrophy
Heterotopia, periventricular
Favism
Hemolytic anemia
Colorblindness, green cone pigment
Incontinentia pigmenti, type II
Hydrocephalus
MASA syndrome
Spastic paraplegia
Rett syndrome
Mature T-cell proliferation
Myopia (Bornholm eye disease)
Mental retardation with psychosis
Endocardial fibroelastosis

Perineal hypospadias
Breast cancer, male, with Reifenstein syndrome
Ectodermal dysplasia, anhidrotic
Alpha-thalassemia/mental retardation
Juberg-Marsidi syndrome
Sutherland-Haan syndrome
Smith-Fineman-Myers syndrome
Hemolytic anemia
Myoglobinuria/hemolysis
Wieacker-Wolff syndrome
Torsion dystonia-parkinsonism, Filipino type
Leukemia, myeloid/lymphoid or mixed-lineage
Anemia, sideroblastic, with ataxia
Allan-Herndon syndrome
Deafness
Choroideremia
Agammaglobulinemia
Fabry disease
Mohr-Tranebjaerg syndrom
Jensen syndrome
Lissencephaly
Bazex syndrome
Mental retardation with growth hormone deficiency
Mental retardation, South African type
Lymphoproliferative syndrome
X inactivation, familial skewed
Pettigrew syndrome
Gustavson mental retardation syndrome
Immunodeficiency, with hyper-IgM
Retinitis pigmentosa
Wood neuroimmunologic syndrome
Heterotaxy, visceral
Albinism-deafness syndrome
Cone dystrophy, progressive
Prostate cancer susceptibility
Fragile X mental retardation
Epidermolysis bullosa, macular type
Diabetes insipidus, nephrogenic
Cancer/testis antigen
Dyskeratosis
Hemophilia A
Hunter syndrome
Mucopolysaccharidosis
Intestinal pseudoobstruction, neuronal
Melanoma antigens
Mental retardation-skeletal dysplasia
Myotubular myopathy
Otopalatodigital syndrome, type I
Colorblindness, red cone pigment
Goeminne TKCR syndrome
Waisman parkinsonism-mental retardation
Barth syndrome
Cardiomyopathy, dilated
Noncompaction of left ventricular myocardium
Von Hippel-Lindau binding protein

Meiosis + human sex chromosomes and X-linked traits

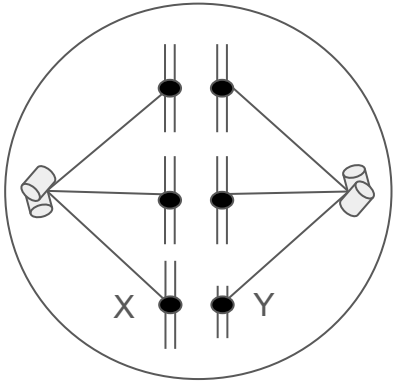
Metaphase I

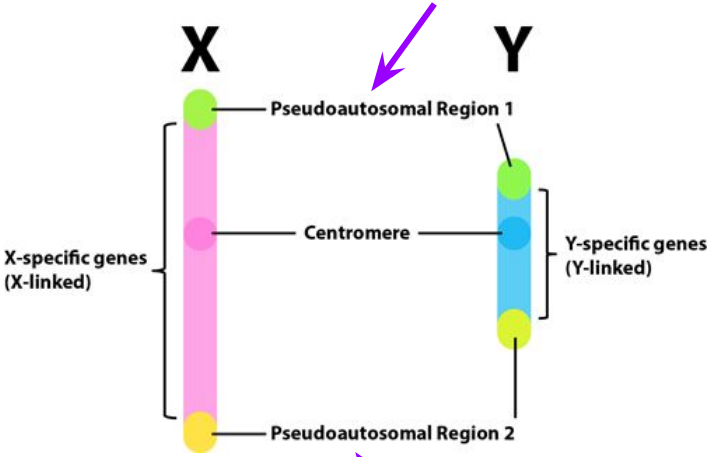
Example from Lecture 6: (Female) germ cell with n=2



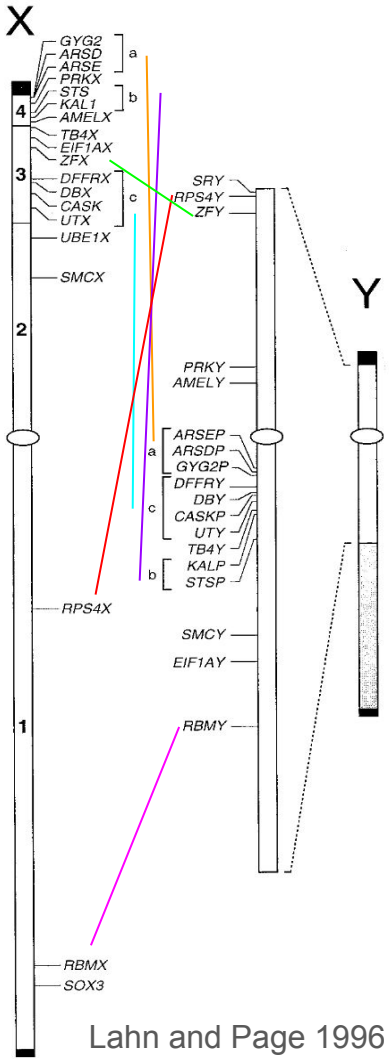
Metaphase plate

Male germ cell with n=3:





X and Y pair at the PAR regions during meiosis I



Lahn and Page 1996

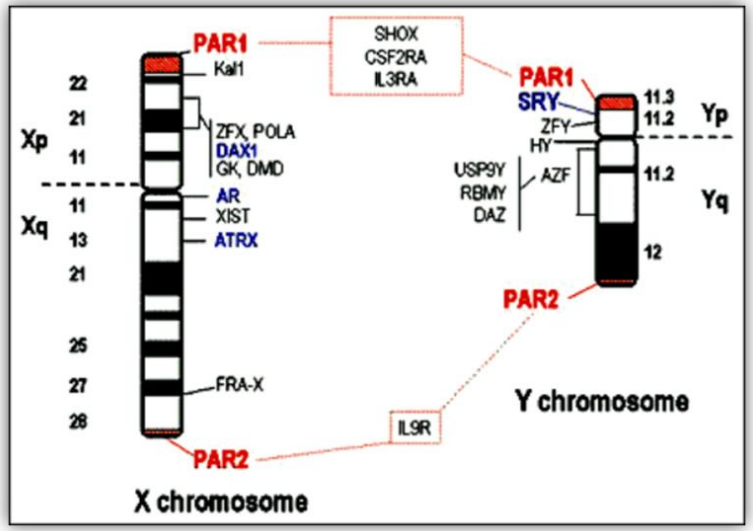
Human sex chromosomes contain sex-determining loci

- Femaleness is an X-linked recessive trait

- Maleness is a Y-linked <u>dominant</u> trait ← the SRY gene triggers male cascade

⇒ <u>How do we know this</u>?

⇒ E.g., what are the phenotypic outcomes of X0, Y0, XXY, and YYX genotypes?

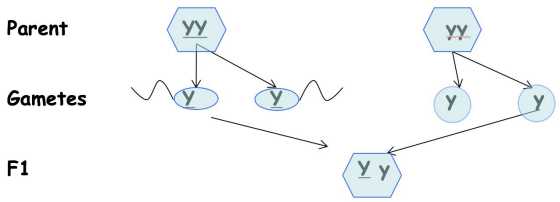


Sex chromosomes contain sex-determining loci

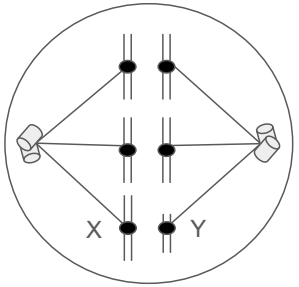
Human genotype-phenotype patterns with X and Y non-disjunction:

XX → Normal females
XY → Normal males

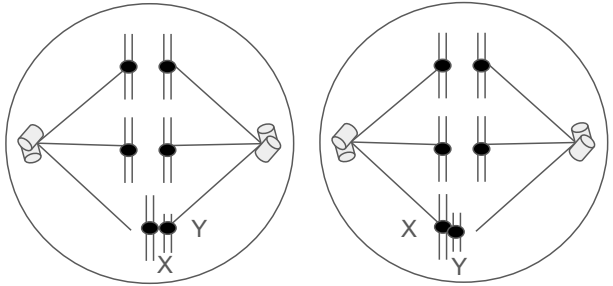
X0 → Females with Turner's Syndrome
Y0 → lethal
XXY → Males with Klinefelter Syndrome



Normal metaphase I of meiosis:



Non-disjunction of X and Y during meiosis I



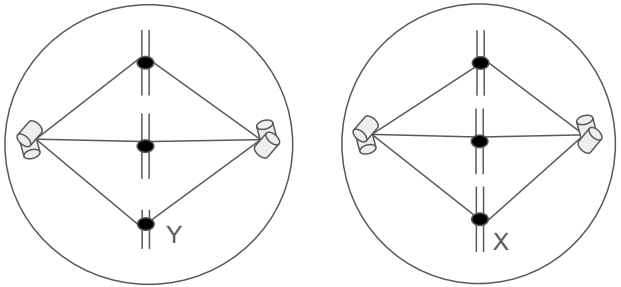
Sex chromosomes contain sex-determining loci

Human genotype-phenotype patterns with X and Y non-disjunction:

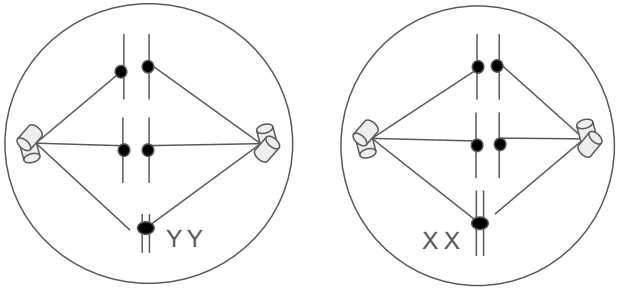
XX → Normal females
XY → Normal males

X0 → Females with Turner's Syndrome
Y0 → lethal
XXY → Males with Klinefelter Syndrome
XYY → Males with Jacobs Syndrome

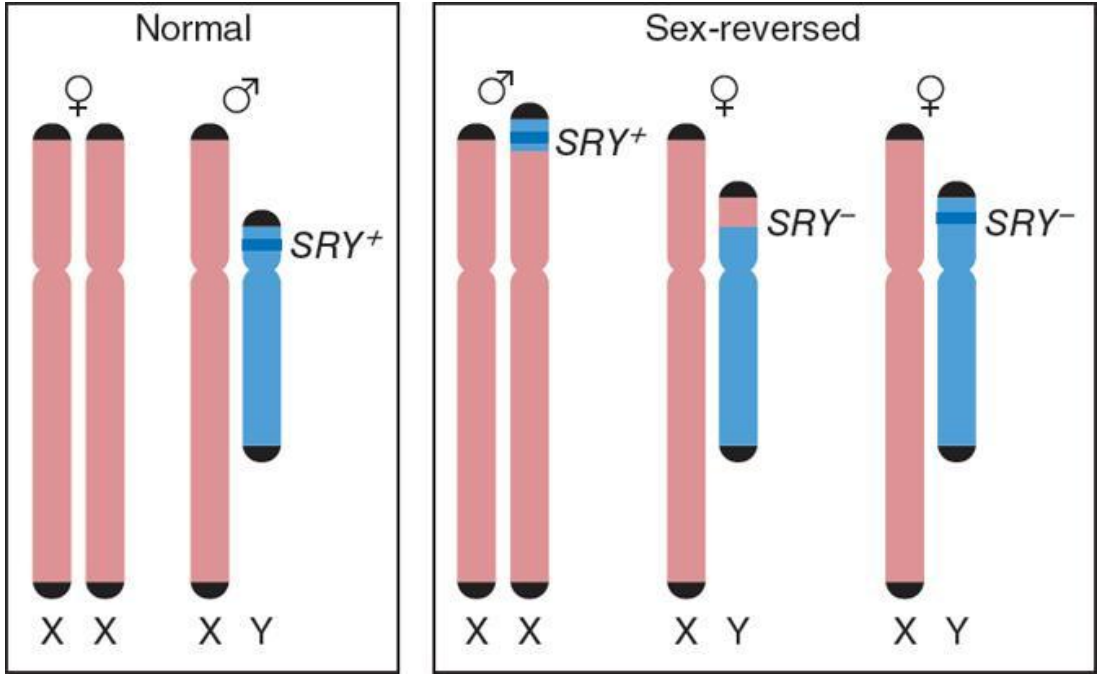
Normal metaphase II of meiosis:



Non-disjunction of X and Y during meiosis II



SRY (sex determining region of Y) gene determines maleness



Sex reversal provided evidence implicating *SRY* as the determinant of maleness

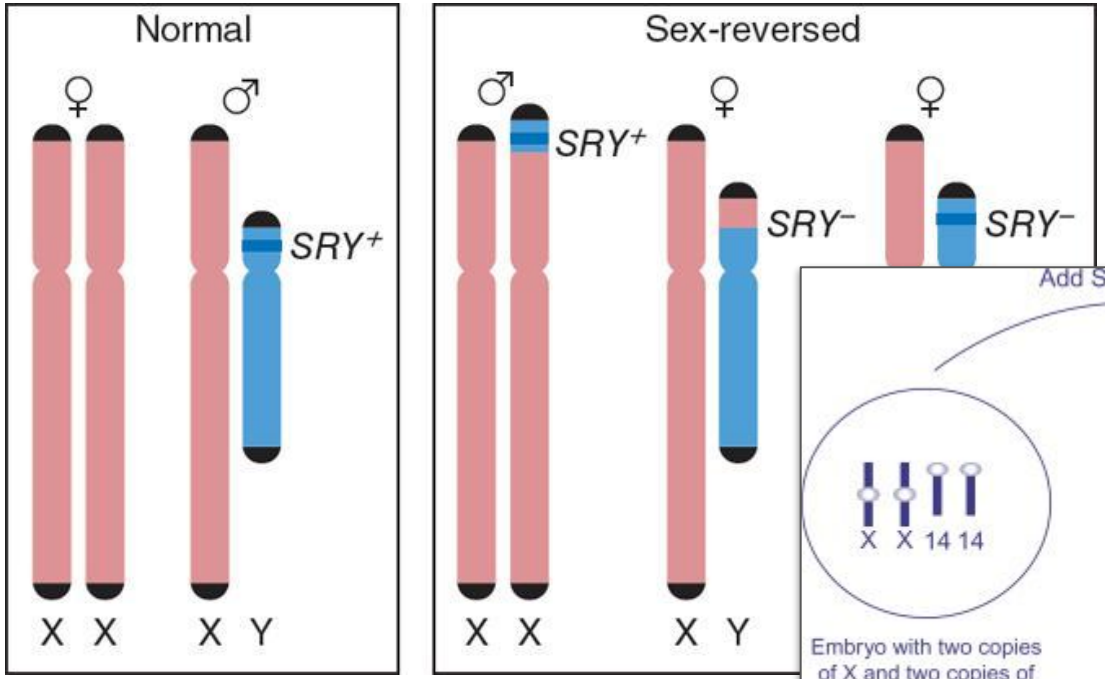
- XX males: SRY locus from Y translocated to X
- XY females: Loss of SRY from Y

XXY → male, Klinefelter syndrome (affects about 1 in 650 newborn males)

XO → No second X, female, Turner Syndrome (1 in 2,500 newborn females)

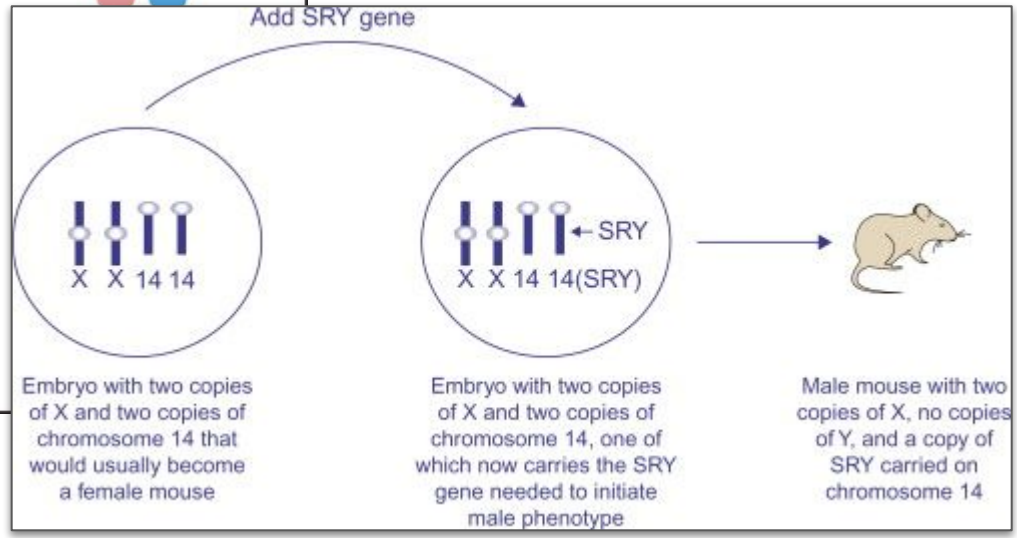
⇒ SRY is necessary and sufficient to invoke the male phenotype in mammals (confirmed experimentally in mice)
⇒ Maleness in mammals is a dominant mendelian trait encoded by SRY

SRY (sex determining region of Y) gene determines maleness



Experimental sex reversal provided evidence implicating SRY as the determinant of maleness

- **XX males: SRY locus from Y translocated to X <u>or an autosome</u>**



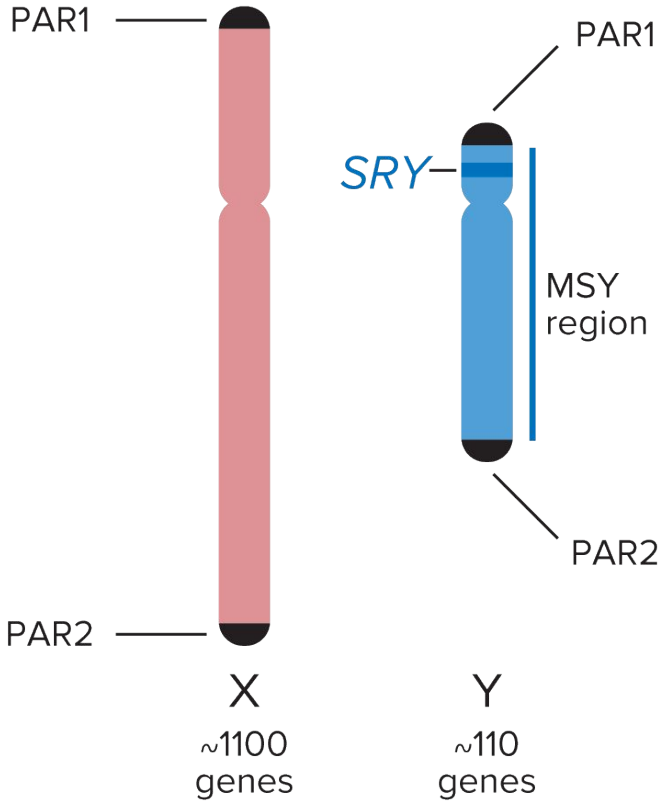
Human sex chromosomes have shared and unique genes

PAR - pseudoautosomal region

- PARs are sites of pairing and recombination, which are necessary for meiosis (particularly for males)
- Contain ~30 genes

MSY - male-specific region

- SRY to activate testis development
- 3 genes for spermatogenesis
- 8 genes with X homologs for dosage (one copy isn't enough)

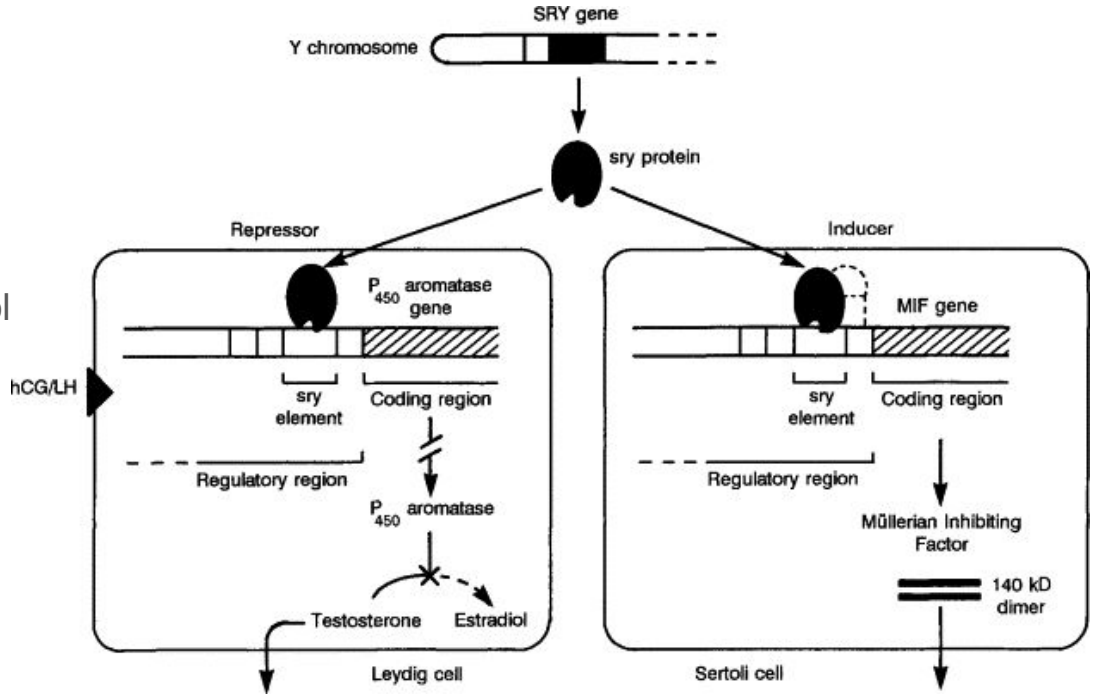


SRY expression sets off a regulatory cascade that blocks female development and starts testosterone expression

SRY protein interacts with promoters of two sex-specific genes:

- Represses: Cytochrome P450 aromatase which mediates the conversion of testosterone to estradiol

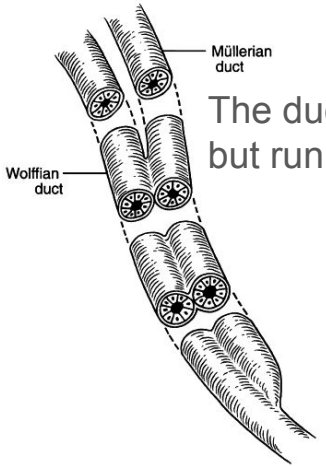
- Induces: Müllerian inhibitory factor (MIF) which mediates Müllerian duct regression, effectively blocking the development of female sex organs.



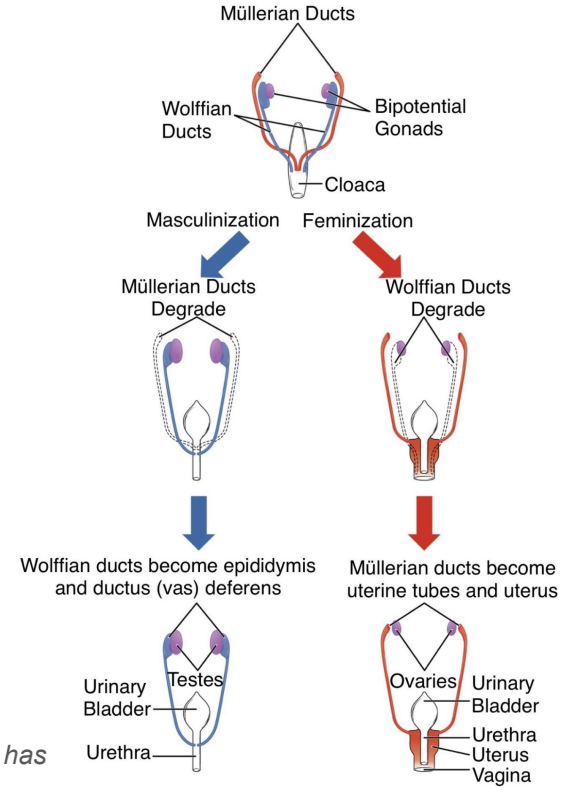
Testosterone conversion blocked (no aromatase)

Ovary development blocked by MIF

SRY expression sets off a regulatory cascade that blocks Müllerian development and enables Wolffian development



The ducts are distinct, but run close together

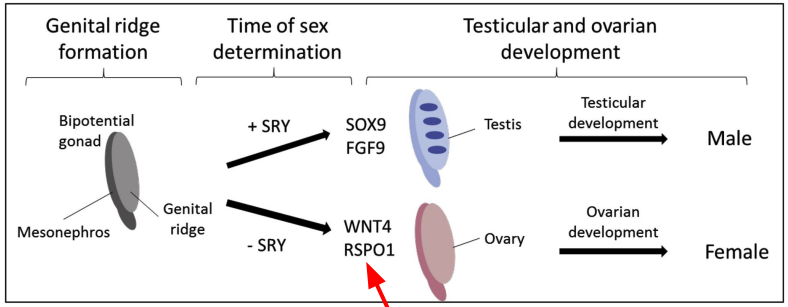


SRY expression
→ Müllerian ducts degenerate
→ Accumulated testosterone stimulates the Wolffian ducts to differentiate into the male internal reproductive tract structures

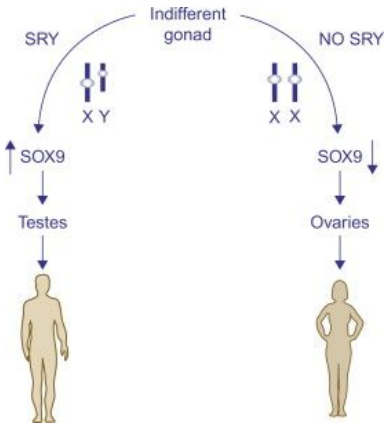
** the central role of the testis in male differentiation has been known since Jost's experiments in the 1940s*

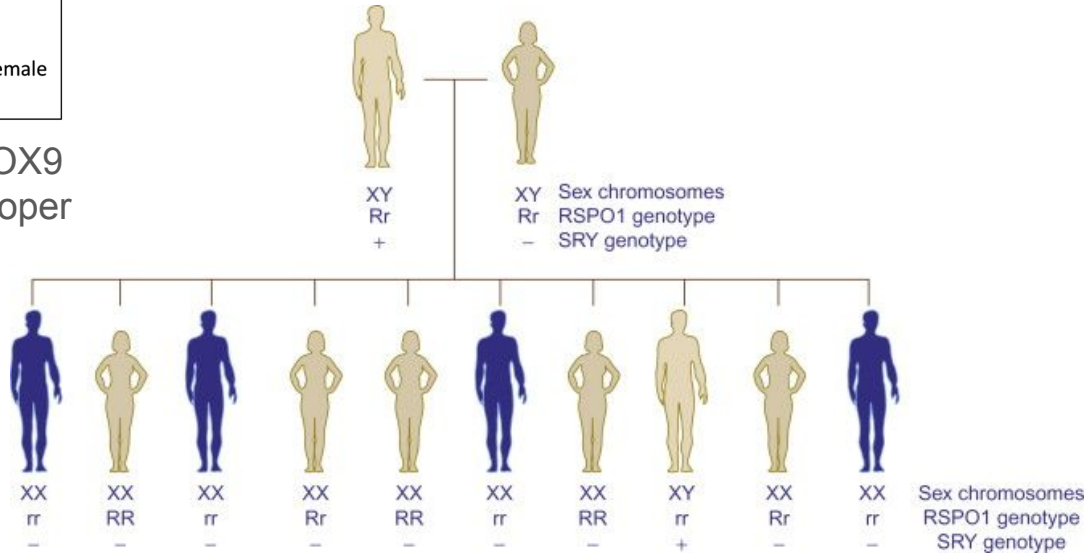
No SRY
→ ovaries differentiate
→ Wolffian ducts degenerate
→ Müllerian ducts develop into a simple columnar epithelial tube that will differentiate into the oviducts, uterus, cervix, and upper portion of the vagina

Importance of knowing the mechanism: a recessive RSPO1 allele induces male differentiation without SRY



RSPO1 represses SOX9 and is required for proper ovarian development





Only SRY male in family

Mechanisms of sex determination differ between species

TABLE 4.2 Mechanisms of Sex Determination

		♀	♂
locus	genetic	**Humans and *Drosophila***	XX	Heterogametic sex → XY
		Moths and *C. elegans*	XX (hermaphrodites in *C. elegans*	XO
		Birds and Butterflies	ZW ← Heterogametic sex	ZZ
ploidy		**Bees and Wasps**	Diploid	Haploid
	environmental	**Lizards and Alligators**	Cool temperature	Warm temperature
		Tortoises and Turtles	Warm temperature	Cool temperature
		Anemone Fish	Older adults	Young adults

Mechanisms of sex determination: fruit flies (Diptera)

Drosophila uses an XY chromosomal system

However, <u>sex is an X-linked dosage-dependent trait in *Drosophila*</u>

- Two Xs confer female development
  - XX and XXY flies are female
  - XY and XO flies are male
- X:A=1

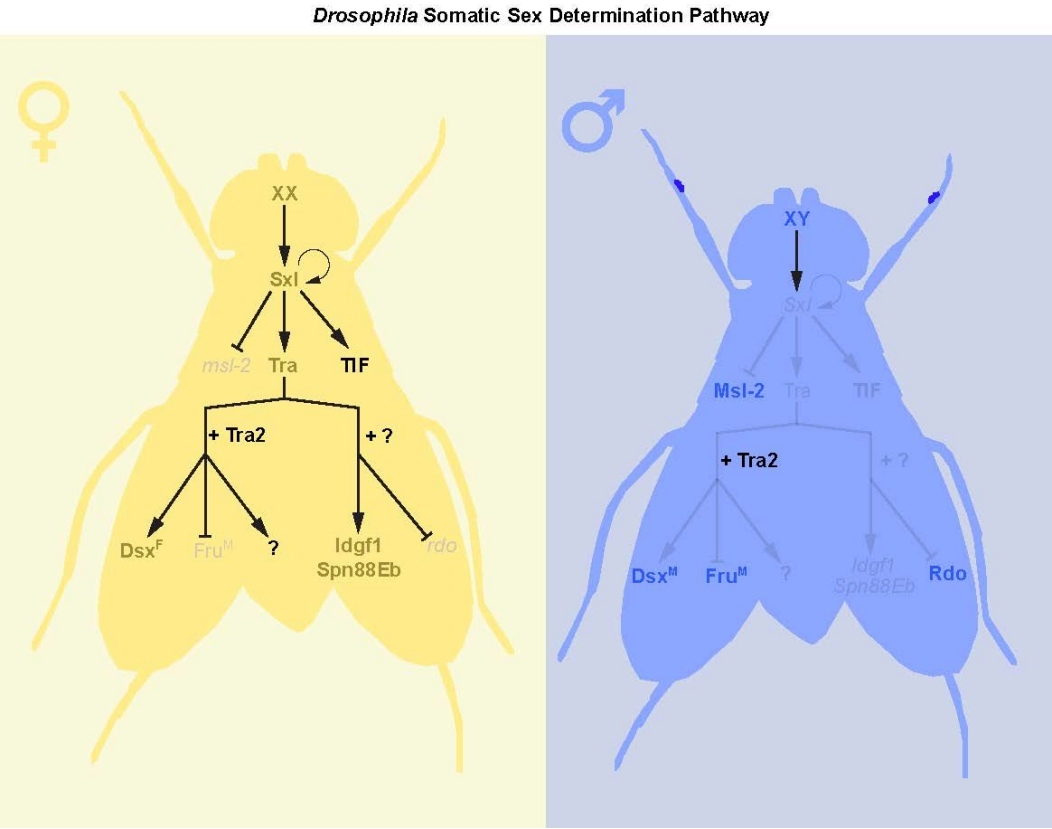
  → Sxl expression triggers differential Dsx^F splicing

- X:A=0.5

  → no Sxl expression. Default to Dsx^M isoform.

The loci on the Y chromosome do not determine sex,

- Required for male fertility
  - XO males are sterile



Sex chromosome gene expression must be modulated in either the hemizygous or homozygous/heterozygous sex

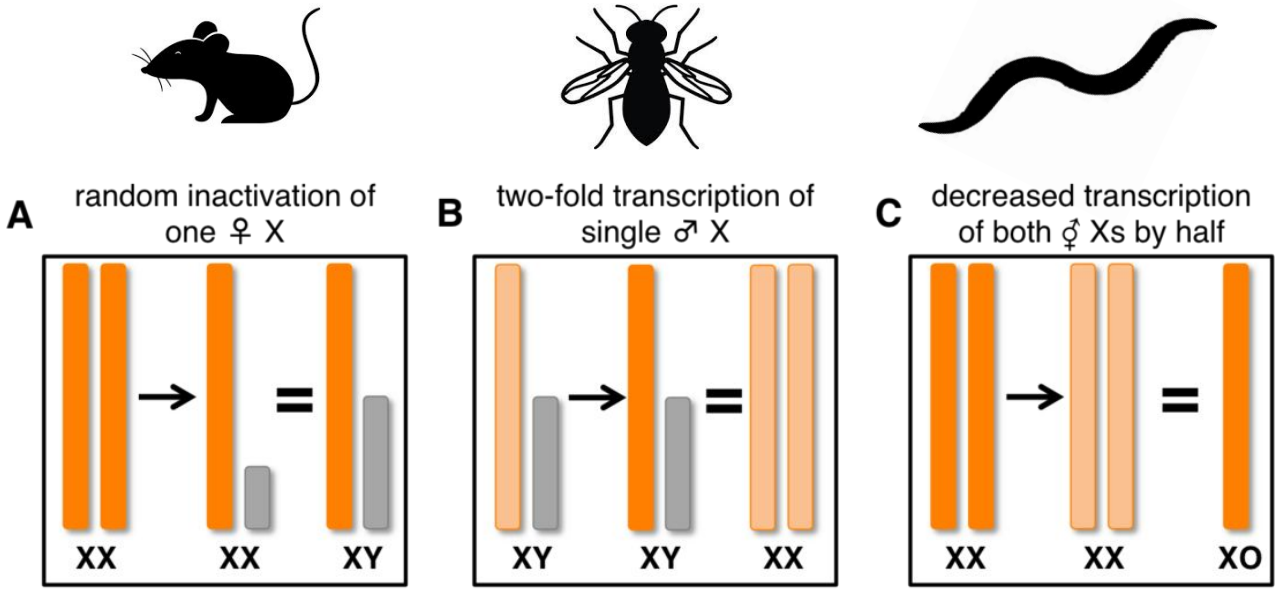


Figure 1. Three main mechanisms of dosage compensation observed in common model eukaryotic organisms.

(A) Mus musculus (as well as humans and many mammalian organisms) follow a pattern of dosage compensation in which females randomly inactive one X chromosome in every cell in their body to match male expression of a single X chromosome.

(B) D. melanogaster follow a mechanism of dosage compensation in which males increase the transcription of their single X chromosome by two-fold in order to match female expression of two X chromosomes.

(C) C. elegans hermaphroditic worms express both X chromosomes by half to match male worms with only one X chromosome.

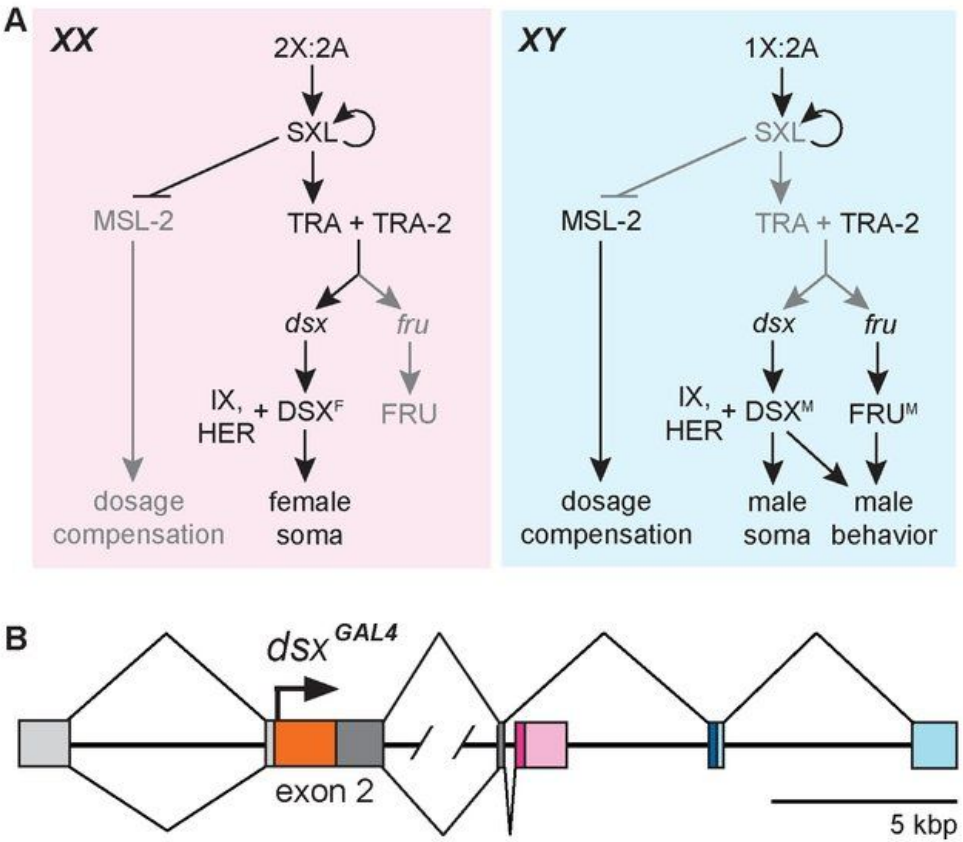
Dosage compensation mechanism in fruit flies

Drosophila uses an XY chromosomal system

However, sex is an X-linked dosage-dependent trait in *Drosophila*

<u>A single X</u> confers maleness and <u>leads to dosage compensation in males</u>

X:A=0.5 → no Sxl expression
→ Default to Dsx^M isoform
→ Default to MSL-2 expression
→ Acetylation of X chromatin
→ Upregulated X expression



Dosage compensation mechanism in mammals

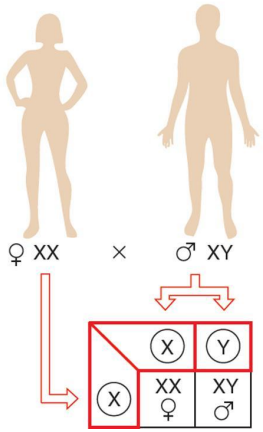
Humans, mice, and other mammals use an XY chromosomal system

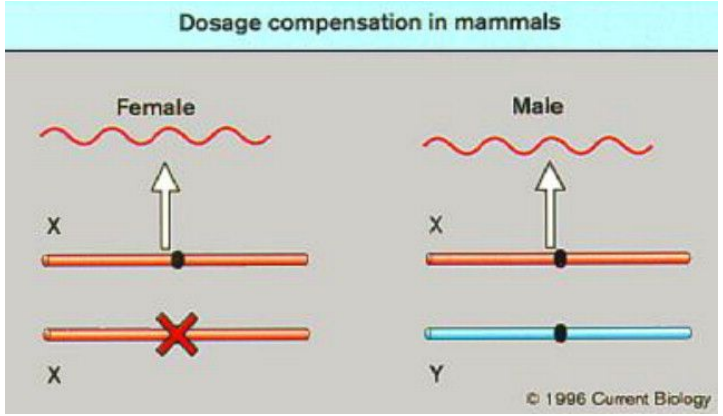
Male development is a Y-linked trait

X chromosome dosage is compensated for in females:

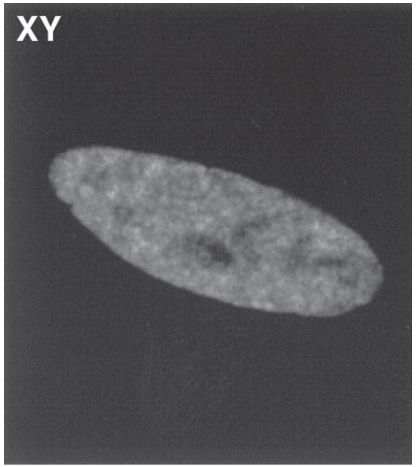
Female mammals randomly inactivate one of their Xs in every cell

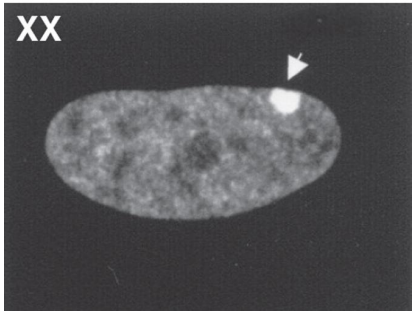
X:Autosomal expression = ½



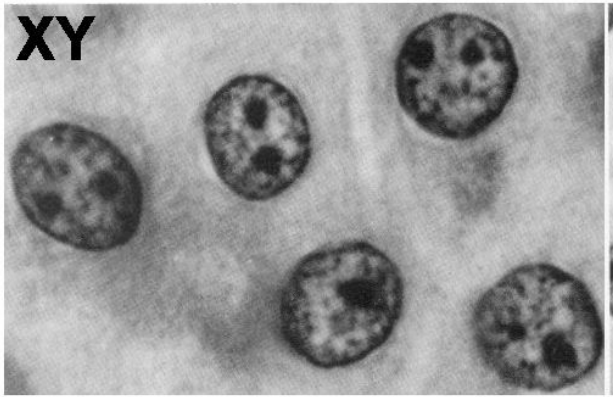


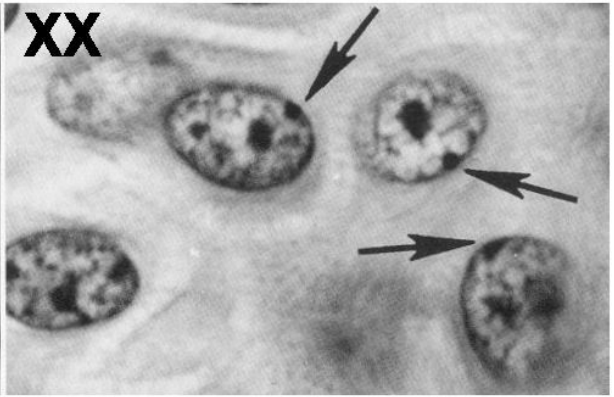
In mammals, the inactive X chromosome is condensed into a Barr body





- One X is randomly selected for inactivation in females
- Inactivation is achieved through chromatin compaction around nucleosomes/histones
- Compacted chromatin, *i.e.*, heterochromatin, is inaccessible for transcription
- Heterochromatin is visibly denser than euchromatin

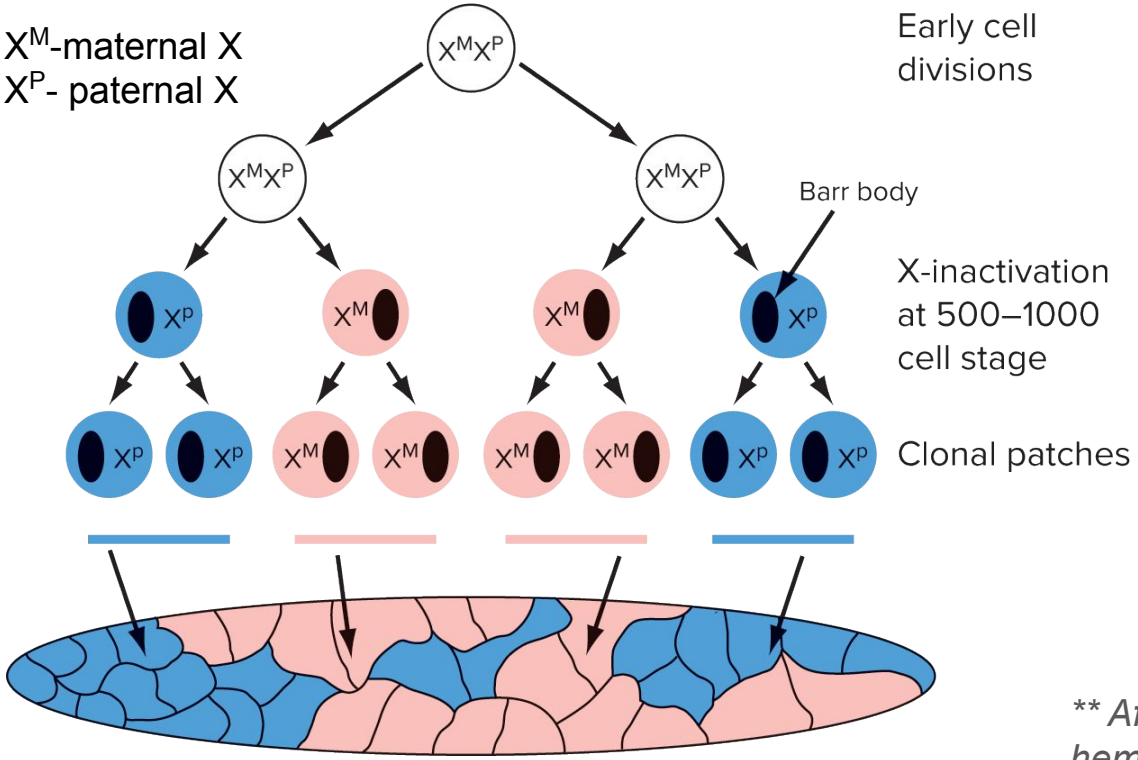




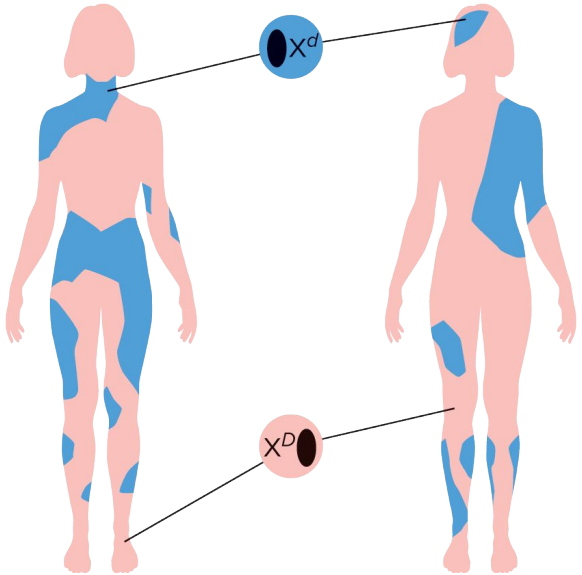
Mammalian females are a patchwork for X-linked gene expression

(a) Perpetuation of X chromosome inactivation after cell divisions

X^M-maternal X
X^P- paternal X

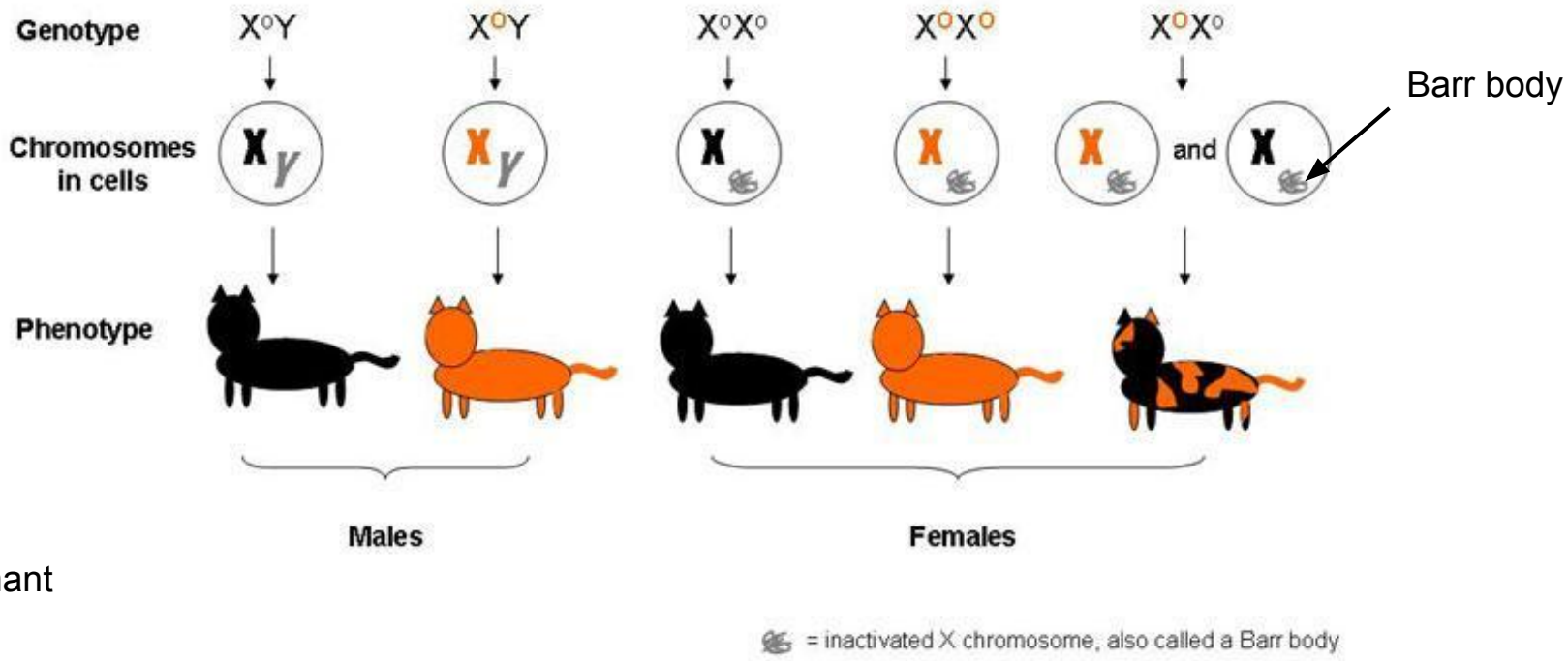


X chromosome inactivation results in patchwork females



*** After inactivation, each cell is functionally hemizygous for X-linked alleles*

Tortoiseshell and Calico female cats are patchworks of X-inactivation



These coat color alleles are codominant (more in Lecture 10)

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X-inactivation is initiated by expression of the *Xist* gene

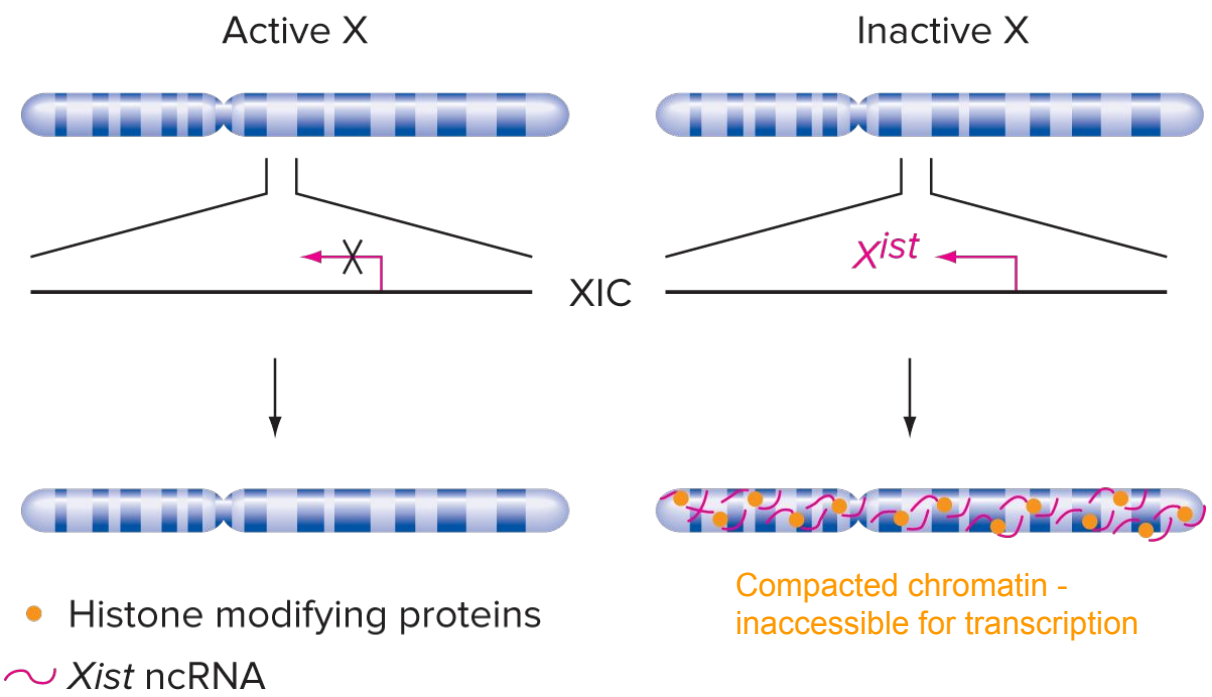
XIC - X inactivation center (450 kb)

- Experiment to show that the XIC causes X-inactivation:
  - A copy of XIC was placed in an autosome and that chromosome becomes a Barr body
- Encodes ***Xist*** RNA - A large (17kb), non-coding, cis-acting regulatory RNA

Xist binds to the X-chromosome that it was expressed from

- Bound *Xist* initiates histone modifications that result in heterochromatin formation
- Still unknown how the inactive X is chosen

(a) *Xist* is transcribed from the inactive X chromosome

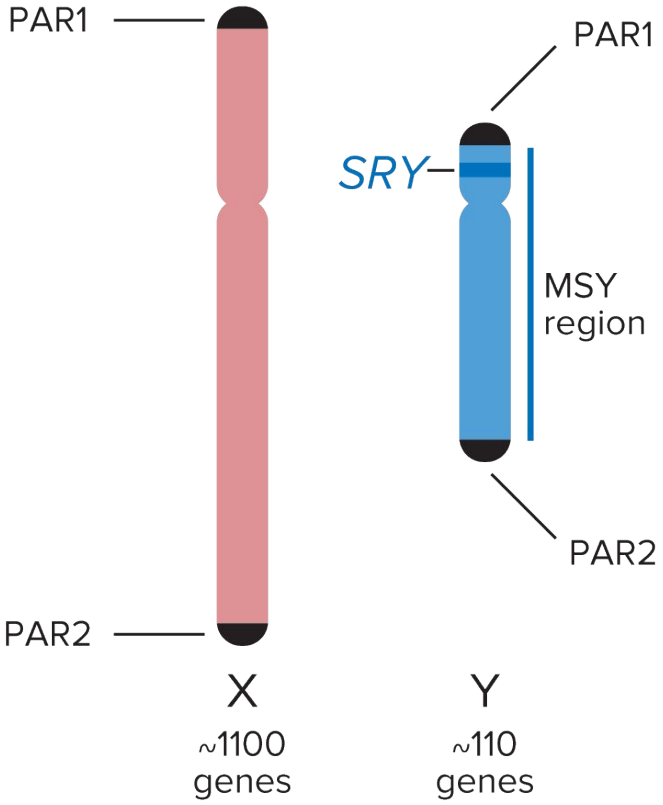


Exceptions to X-inactivation

- X-linked genes with homologous copies on the Y
  - 30 PAR genes
  - 8 genes in MSY

- Klinefelter males - XXY
  - One X is inactivated
  - Still express 3x doses of X/Y homologs

- Turner females - X0
  - No X inactivation
  - Only 1x dose of X/Y homologs



Differential phenotypic outcomes of sex chromosomal aneuploidy among species are explained by sex determination and dosage compensation mechanisms

TABLE 4.1 Sex Determination in Fruit Flies and Humans

	Complement of Sex Chromosomes						
	XXX	**XX**	**XXY**	**XO**	**XY**	**XYY**	**OY**
Drosophila	Dies	Normal female	Normal female	Sterile male	Normal male	Normal male	Dies
Humans	Nearly normal female	Normal female	Klinefelter male (sterile); tall, thin	Turner female (sterile); short, webbed neck	Normal male	Normal or nearly normal male	Dies

Humans can tolerate extra X chromosomes (e.g., XXX) better than can *Drosophila* because in humans all but one X chromosome becomes a Barr body, as discussed later in this chapter. Complete absence of an X chromosome is lethal to both fruit flies and humans. Additional Y chromosomes have little effect in either species. Although the Y chromosome in *Drosophila* does not determine whether a fly looks like a male, it is necessary for male fertility; XO flies are thus sterile males.